8 (a) (i) The Moon is continuously attracted towards the Earth by gravitational force and orbits around the Earth.

Explain why the Moon does not fall into the Earth.

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..... [2]

(ii) Assuming Earth to be a perfect sphere, explain why the weight of a person measured on an electronic balance at the poles is greater than that measured at the equator.

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..... [4]

(b) The Moon has an orbital period of about 27.3 days. The mass of the Earth is 6.00×10^{24} kg and the mass of the Moon is 7.34×10^{22} kg.

(i) Show that the distance between the centres of the Moon and the Earth is approximately 383 000 km.

[3]

(ii) Hence, calculate the distance *x*, from the centre of the Earth along the Earth-Moon line to the point Q where the resultant gravitational field strength due to the Earth and Moon is zero.

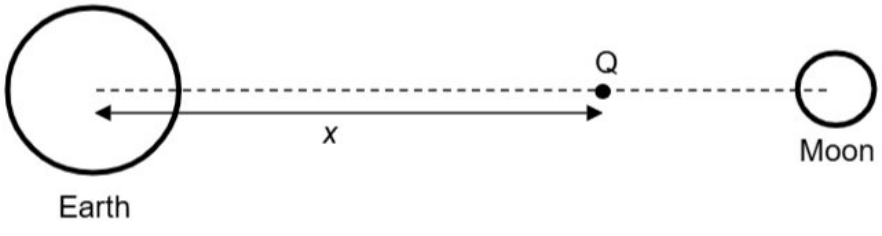


Figure 8.1 (not to scale)

x = ...m [2]

(iii) Calculate the gravitational potential at point Q.

gravitational potential at point Q = ……J kg^{-1} [2]

(iv) The radius of the Earth is 6370 km and the radius of the Moon is 1740 km.

Determine the minimum energy required to send a stationary capsule of mass 30 kg from the surface of the Earth to the surface of the Moon.

minimum energy = ……J [4]

(c) A planet of radius R is isolated in space. It has no atmosphere, and its mass can be assumed to be concentrated at its centre.

A projectile is launched from the surface of the planet with a **minimum** kinetic energy ***E*** required for the projectile to **not** return to the planet.

On Figure 8.2, draw:

(i) a line, labelled **P**, to show how the gravitational potential energy of the projectile varies with distance from the centre of the planet as it moves from R to $4R$,

(ii) a line, labelled **K**, to show how the kinetic energy of the projectile varies with distance from the centre of the planet as it moves from R to $4R$.

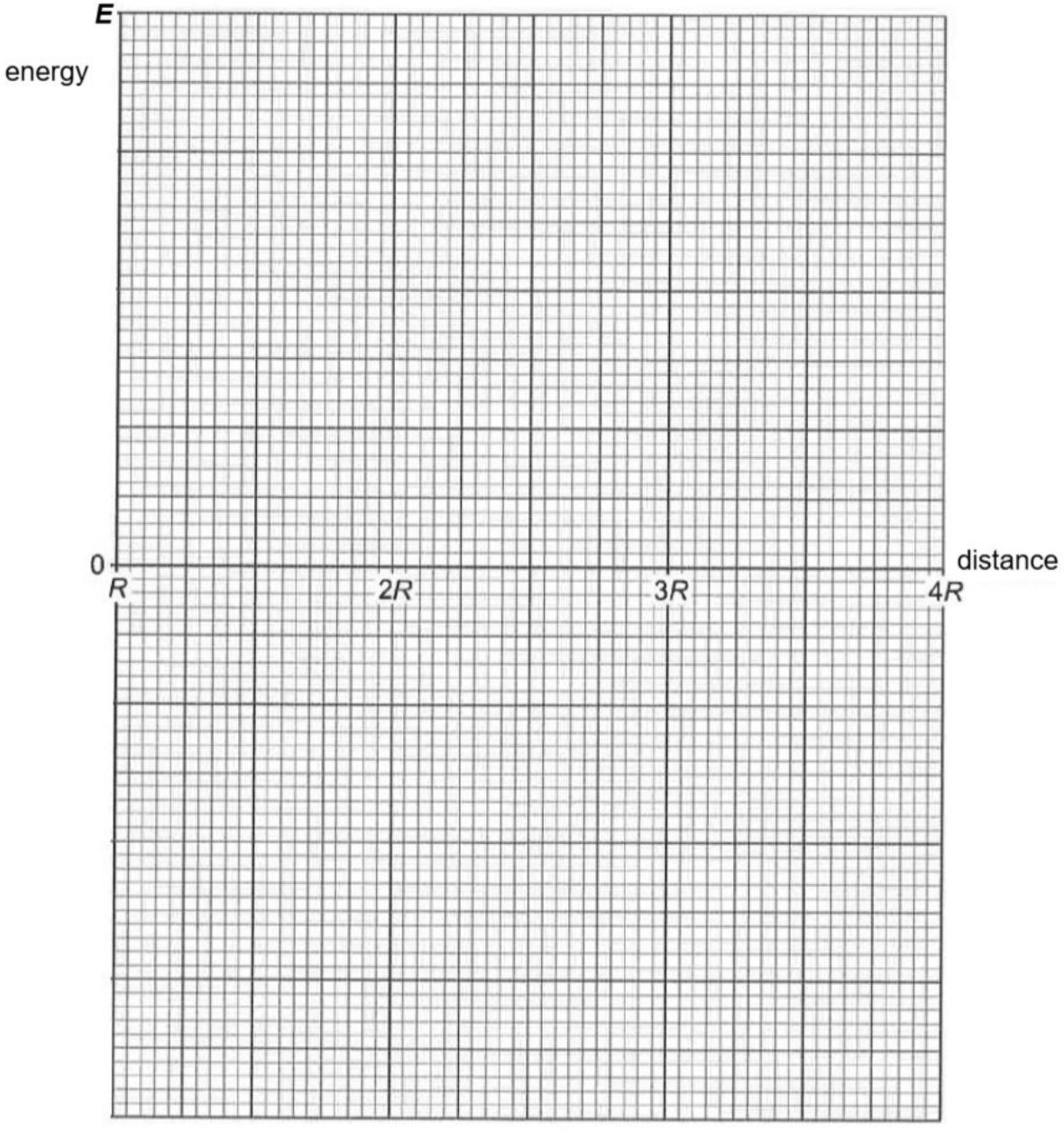


Figure 8.2

[3]

[Total: 20]